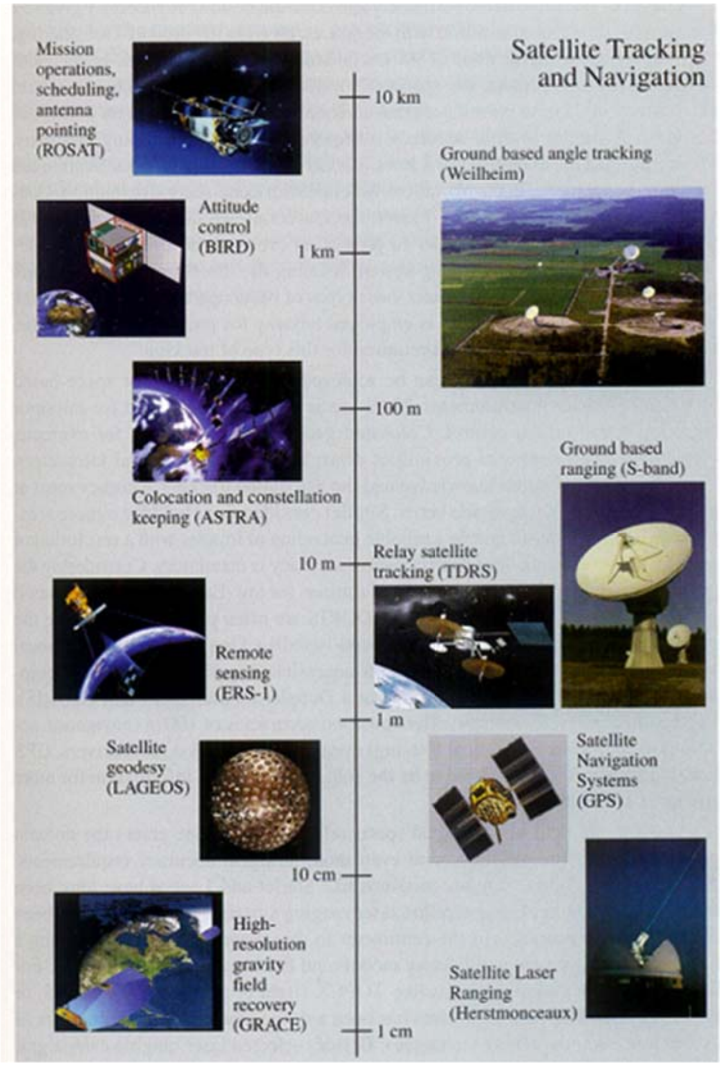
Chapter 4. Observations

4.1. Introduction

- Science/Astrodynamics development consists in
  - Ideas/Theoretical Explanations
  - Practice: Observation and Computation
- In this Chapter (and later), we simply follow the footsteps to Tycho Brahe and Kepler with modern sense and technology to DETERMINE ORBIT



4.2. Obtaining Data

- We derive all data (range, range rate, azimuth, elevation) from signals of electromagnetic radiation
  - $\lambda = c / f$

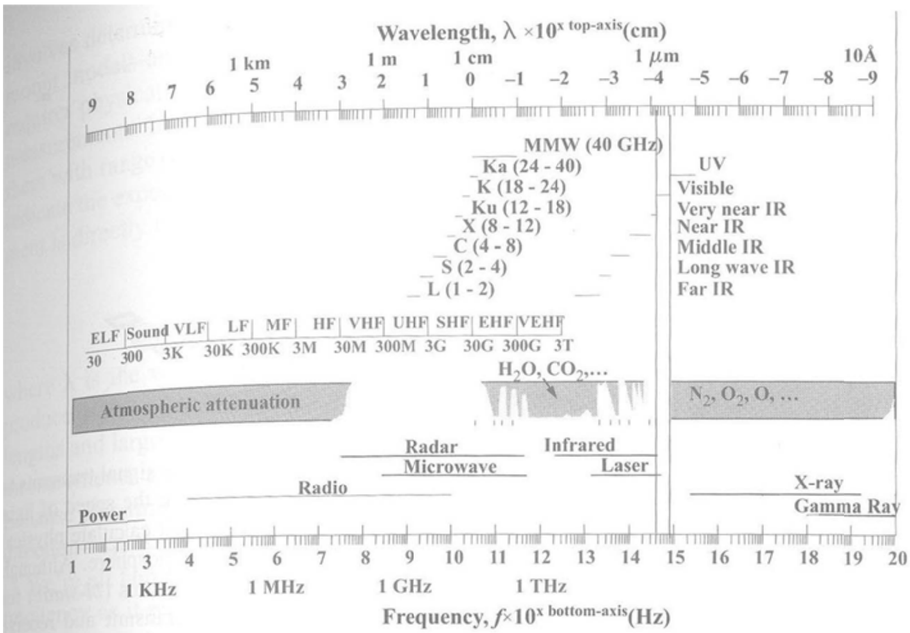


Figure 4-1. **Electromagnetic Spectrum.** Atmospheric attenuation partially blocks much of the incoming electromagnetic radiation, as the dark bar illustrates. Frequency values are given for various bands next to each letter descriptor. Values are in GHz, THz, and so on, as appropriate. [Hovanessian (1988:5–7), Chetty (1988:311), Gammon (1976:17), Morgen & Gordon (1989:233)]

- Atmospheric and Timing variations tend to degrade the performance of observations
  - Knowledge of Atmosphere is not complete
    - Ionosphere/Troposphere/Rain and Fog cause Slowness and Distortion
  - Observations usually reference UTC as the standard time
    - Difficult to implement UTC at an observation site exactly
      - On-site Atomic Clocks/Broadcast UTC/GPS signal
    - Disseminate UTC to all local sensors can be challenging

- Sensors receiving Signal
  - One way vs. Two way (Reflected/Retransmitted, Monostatic/Bistatic)

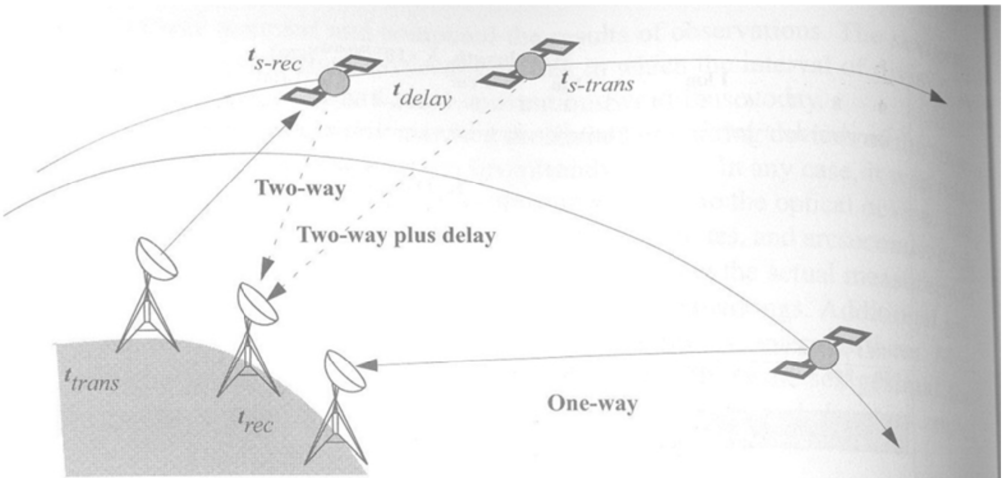


Figure 4-2. Geometry for One-way and Two-way Transmission. When a signal transmits to or from a satellite, we determine precise values of range from the speed of light and the total time (including any delays) for the trip. We must calculate physical and atmospheric distortions during each trip through the atmosphere. Although I've shown the two-way transmissions using two separate stations (***bi-static***) for clarity, the most common case is for the same station to transmit and receive (***mono-static***).

  - Accuracy
    - Radar Measurements (~3m)
    - Laser (~1mm)
    - Refer to Figure 4-1 for typical wavelengths

- Figure 4-2 suggests that we need to have 3 measurements
  - Total Time (of Electromagnetic Waves Travel): Δt
  - Direction of Signal: θ_{BW}
  - Doppler Shift: f_d

- The ultimate goal of most forms of tracking
  - Range

$$\rho = \frac{c\Delta t}{2}$$
$$\rho = \left| r_{sat}(t_{s-rec}) - r_{site}(t_{trans}) \right| + \left| r_{site}(t_{rec}) - r_{sat}(t_{s-trans}) \right|$$

  - Direction of Signal
    - Signal itself + Physical Sensor
    - Less accurate than Range and Range Rate in general (due to physical measurements)
    - Less sensitive (due to scale issue, i.e., $r\Delta\theta$)
    - Accuracy of angular measurement is directly related to beamwidth

$$\theta_{BW} = \frac{\lambda}{D}, \qquad D : \text{aperture diameter}$$

  - Range Rate
    - Doppler Shift f_d in $f_0 + f_d$

$$f_d = -\frac{2\dot{\rho}}{\lambda}$$

    - Only holds for system where signal frequency does not change due to retransmission

4.2.1. Quantity of Data

- In general, the more the data, the better, especially for initial orbit determination for which we have little/no knowledge
  - Short-Arc

- Observations limited to small arc
- Often better indicators of near-term satellite motion

- Long-Arc
  - Observations over multiple revolutions
  - Usually preferred, as it provides more accurate determination of the orbit

4.2.2. Types of Data

- Seven combinations of basic data types

TABLE 4-1. Deciding on an Appropriate Technique for Initial Orbit Determination. This table pairs solution techniques with available observational data. Each data type must include station, satellite location (ϕ_{gd}, λ, h_{ellp}) and time (UTC). Radars provide most of this data (ρ, β, el, $\dot{\rho}$). Lasers yield range data, and optical devices produce α_t and δ_t. Sec. 7.4.1 discusses techniques which can process ρ and $\dot{\rho}$ data.

Data Type	Restrictions	Sec./Chap.	Solution Method
$\dot{\rho}$	None	10	Estimation
β, el	3 sets minimum	7.3	*LAPLACE, GAUSS, DOUBLE-R*
ρ, β, el	2 sets minimum	7.2, 7.5, 7.6	*SITE-TRACK*, then *LAMBERT* (2) or *GIBBS/HGIBBS* (3)
ρ, β, el, $\dot{\rho}$	2 or 3 sets minimum	7.2	*SITE-TRACK*
ρ, β, el, $\dot{\rho}$, $\dot{\beta}$, $\dot{el}$	None	7.2	*SITE-TRACK*
α_t, δ_t	3 sets minimum	7.3	*LAPLACE, GAUSS, DOUBLE-R*
ρ	6 simultaneous, None	7.4, 10	Trilateration, Estimation

  - $\dot{\rho}$: range rate
  - β, el: azimuth, elevation
  - α_t, δ_t: topocentric right ascension, declination
  - ρ: range

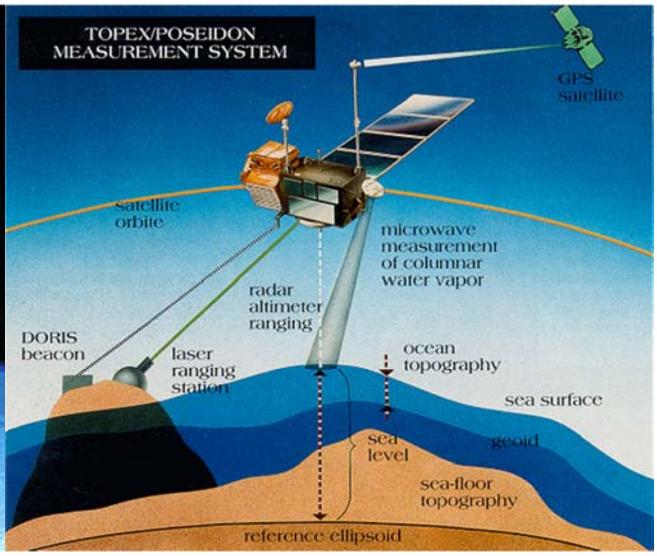
- Each data type must include
  - Station Information
  - Satellite Location $(\phi_{gd}, \lambda, h_{ellp})$
  - Time (UTC)
- Radars provide $(\rho, \beta, el, \dot{\rho})$

  Lasers provide ρ

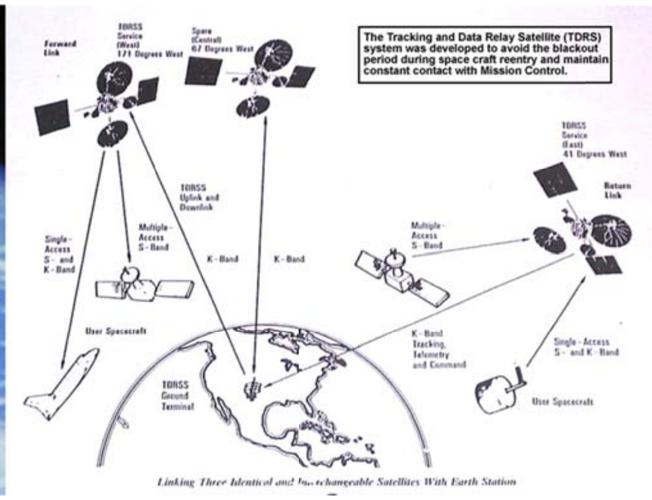
  Optical Devices provide (α_t, δ_t)

4.2.3. Example Applications

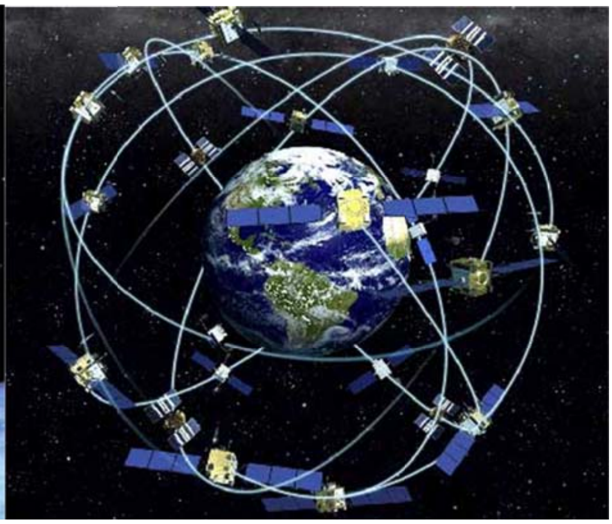
- TOPEX/Poseidon
  - Contains 5 systems for Orbit Determination (OD)
    - Laser Retro-Reflectors
    - Dual Doppler receiver for DORIS
    - GPS Demonstrations Receivers (GPSDR)
    - TDRS (Tracking and Data Relay Satellite)
    - Single and Dual Channel Altimeters
  - Useful for Exploring Orbit Determination techniques
  - 2cm radial accuracy is typically achieved
  - Post-processed solution



- NASA TDRS (Tracking and Data Relay Satellite)
  - Similar techniques to those of TOPEX
  - Bilateration Ranging Transponder system provides main tracking capability



- GPS
  - Transmits signals to be processed into range and timing information
  - LEO orbit <2~3meters accuracy

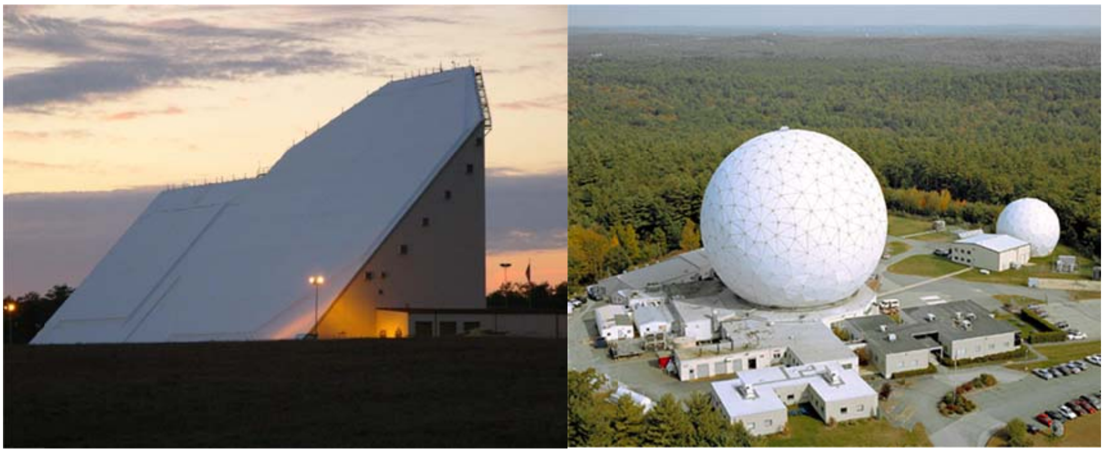


4.3. Introduction to Sensor Systems

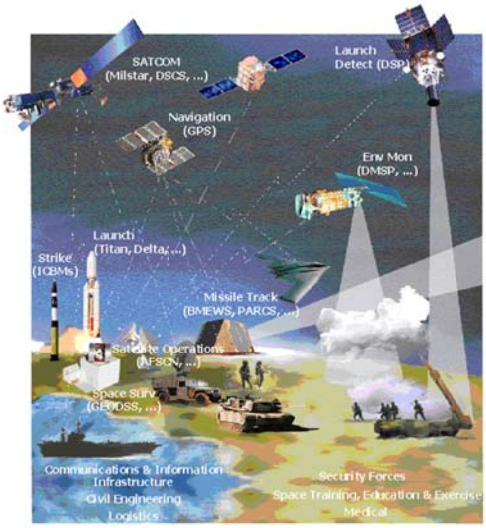
- Space Surveillance: General practice of observing objects in orbit

- Sensor Categories
  - Dedicated sensors perform primary mission such as space surveillance, missile warning
  - Collateral sensors perform other primary missions, but still support space surveillance
  - Contributing sensors include most university, commercial, and other sensors

- Tracking Network: Globally distributed sensors to acquire data on satellites
  - US Space Surveillance Network (SSN)



  - Russian Space Surveillance System (RSSS)
  - Air Force Satellite Control Network (AFSCN): most sensors are dedicated



- Deep Space Network (DSN): managed by JPL

- Satellite Laser Ranging Network (SLRN)
    - International consortium of 50 dedicated/contributing sensors
    - ~1cm range accuracy
    - Use calibration satellites such as LAGEOS

- DORIS system
  - Track TOPEX/Poseidon SPOT-4

- Tracking data is imperfect and contains noises and biases
  - Calibration is required

4.4. Observation Transformations

- Correct use of observations requires us to combine
  - Location Parameters
  - Coordinate System and Angular Measurements
  - Motion of Coordinate Systems
  - Corrections to Observations

- Then, we must correctly process the coordinate system information
  - Site Location
  - Observations
  - Any Reference Data (e.g., Star Catalog)
  - Other Necessary Systems for further processing

- We care the conversion between:
  - Angles ⇔ Position/Velocity vector
  - Angles of Different Systems

4.4.1. Geocentric Right Ascension & Declination

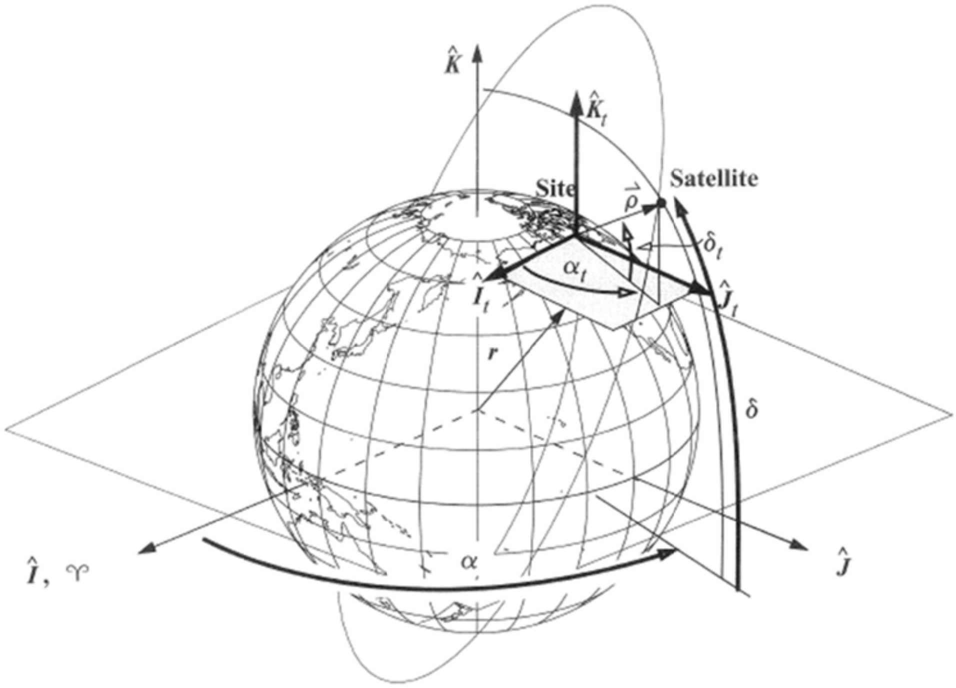


Figure 4-5. **Right Ascension and Declination.** Geocentric right ascension and declination use the Earth's equatorial plane. Topocentric right ascension and declination values use a plane parallel to the Earth's equator but located at a particular site. Transformations rely on determining the slant-range vector ($\hat{\rho}$) in the IJK system, and the radius vector to the satellite, *r*.

- Algorithm 25: $\left(\vec{r}_{IJK}, \vec{v}_{IJK}\right) \Rightarrow \left(r, \alpha, \delta, \quad \dot{r}, \dot{\alpha}, \dot{\delta}\right)$

  - $r = \left|\vec{r}_{IJK}\right|$

  - $\sin\delta = \frac{r_K}{r} \quad , \quad \cos\delta = \frac{\sqrt{{r_I}^2 + {r_J}^2}}{r}$

  - If $\sqrt{r_I^2 + r_J^2} \neq 0$

$$\sin\alpha = \frac{r_J}{\sqrt{r_I^2 + r_J^2}} \quad , \quad \cos\alpha = \frac{r_I}{\sqrt{r_I^2 + r_J^2}}$$

$Else\ \sqrt{r_I^2 + r_J^2} = 0$

$$\sin\alpha = \frac{v_J}{\sqrt{v_I^2 + v_J^2}} \quad , \quad \cos\alpha = \frac{v_I}{\sqrt{v_I^2 + v_J^2}}$$

  - From $\vec{r} \cdot \vec{v} = r\dot{r}$

$$\dot{r} = \frac{\vec{r} \cdot \vec{v}}{r}$$

$$\vec{r}_{IJK} = \begin{bmatrix} r\cos\delta\cos\alpha \\ r\cos\delta\sin\alpha \\ r\sin\delta \end{bmatrix}$$

$$\vec{v}_{IJK} = \begin{bmatrix} \dot{r}\cos\delta\cos\alpha - r\sin\delta\cos\alpha\ \dot{\delta} - r\cos\delta\sin\alpha\ \dot{\alpha} \\ \dot{r}\cos\delta\sin\alpha - r\sin\delta\sin\alpha\ \dot{\delta} + r\cos\delta\cos\alpha\ \dot{\alpha} \\ \dot{r}\sin\delta + r\cos\delta\ \dot{\delta} \end{bmatrix}$$

- From the v_K − component

$$\dot{\delta} = \frac{v_K - \dot{r}\sin\delta}{r\cos\delta}$$

$$\Rightarrow \dot{\delta} = \frac{v_K - \dot{r}\dfrac{r_K}{r}}{\sqrt{r_I^{\,2} + r_J^{\,2}}}$$

- $-\left(r \times v\right)_K = r_J v_I - r_I v_J = \cdots = -\dot{\alpha}\left(r_I^{\,2} + r_J^{\,2}\right)$

$$\Rightarrow \dot{\alpha} = \frac{r_I v_J - r_J v_I}{\left(r_I^{\,2} + r_J^{\,2}\right)}$$

4.4.2. Topocentric Right Ascension & Declination

- Observations usually come from a particular site, and are thus in the topocentric frame

$$\vec{r}_{IJk} = \vec{\rho}_{IJK} + \vec{r}_{SiteIJK}$$
$$\vec{v}_{IJK} = \dot{\vec{\rho}}_{IJK} + \vec{v}_{SiteIJK}$$

- Algorithm 26: $\left(\vec{r}_{IJk}, \vec{v}_{IJK}, \vec{r}_{SiteIJK}, \vec{v}_{SiteIJK}\right) \Rightarrow \left(\rho, \alpha_t, \delta_t, \dot{\rho}, \dot{\alpha}_t, \dot{\delta}_t\right)$

$\Leftrightarrow$Translation + Algorithm 25

$$\begin{cases} \vec{\rho}_{IJK} = \vec{r}_{IJk} - \vec{r}_{SiteIJK} \\ \dot{\vec{\rho}}_{IJK} = \vec{v}_{IJK} - \vec{v}_{SiteIJK} \end{cases}$$

- $\rho = \left|\vec{\rho}_{IJK}\right|$

- $\sin\delta_t = \dfrac{\rho_K}{\rho}$

- If $\sqrt{\rho_I^2 + \rho_J^2} \neq 0$

$$\sin\alpha_t = \frac{\rho_J}{\sqrt{\rho_I^2 + \rho_J^2}} \quad , \quad \cos\alpha_t = \frac{\rho_I}{\sqrt{\rho_I^2 + \rho_J^2}}$$

  Else $\sqrt{\rho_I^2 + \rho_J^2} = 0$

$$\sin\alpha_t = \frac{\dot{\rho}_J}{\sqrt{\dot{\rho}_I^2 + \dot{\rho}_J^2}} \quad , \quad \cos\alpha_t = \frac{\dot{\rho}_I}{\sqrt{\dot{\rho}_I^2 + \dot{\rho}_J^2}}$$

- $\dot{\rho} = \dfrac{\vec{\rho}_{IJK} \cdot \dot{\vec{\rho}}_{IJK}}{\rho}$

- $\dot{\delta}_t = \dfrac{\dot{\rho}_k - \dot{\rho}\dfrac{\rho_K}{\rho}}{\sqrt{\rho_I{}^2 + \rho_J{}^2}}$

- $\dot{\alpha}_t = \dfrac{\rho_I \dot{\rho}_J - \rho_J \dot{\rho}_I}{\rho_I{}^2 + \rho_J{}^2}$

4.4.3. Azimuth-Elevation

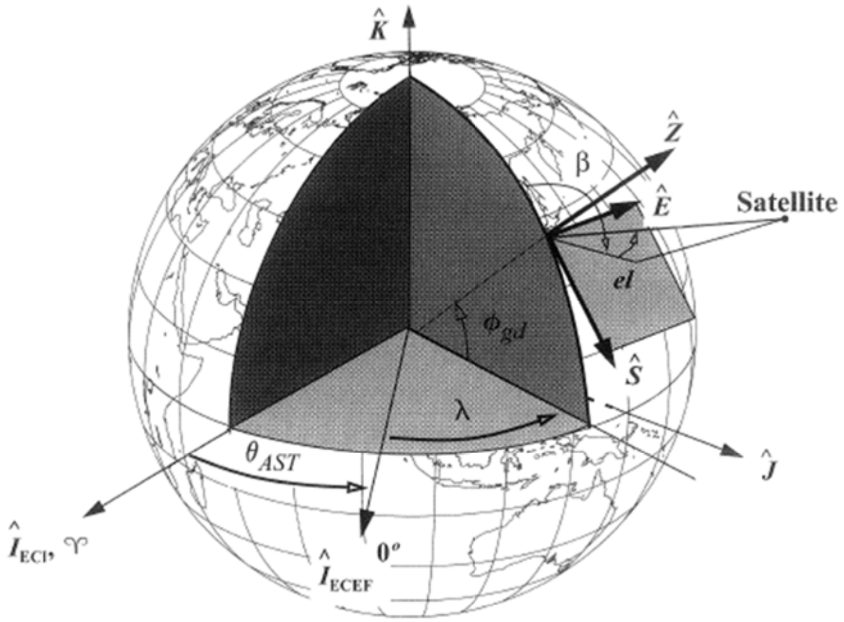


Figure 4-6. Topocentric-Horizon Coordinate System (SEZ). We measure azimuth, β, and elevation, *el*, in the topocentric-horizon coordinate system. Most analyses consider the differences between ECI and ECEF coordinate systems.

- Position/Velocity vectors → $\left(\rho, \beta, el,\ \dot{\rho}, \dot{\beta}, \dot{el}\right)$ mainly applies to orbit determination and estimation process

- Algorithm 27 : $\left(\vec{r}_{IJK}, \vec{v}_{IJK}, \vec{r}_{siteIJK}, \phi_{gd}, \theta_{LST}\right) \Rightarrow \left(\rho, \beta, el,\ \dot{\rho}, \dot{\beta}, \dot{el}\right)$

$$\vec{r}_{IJK} = \vec{\rho}_{IJK} + \vec{r}_{siteIJK} \quad \Rightarrow \vec{\rho}_{IJK} = \vec{r}_{IJK} - \vec{r}_{siteIJK}$$

$$\left.\begin{aligned} \vec{v}_{IJK} &= \dot{\vec{\rho}}_{IJK} + \vec{v}_{siteIJK} \\ &= \dot{\vec{\rho}}_{IJK} + \vec{\omega}_{\oplus} \times \vec{r}_{siteIJK} \end{aligned}\right) \Rightarrow \dot{\vec{\rho}}_{IJK} = \vec{v}_{IJK} - \vec{\omega}_{\oplus} \times \vec{r}_{siteIJK}$$

$$\vec{\rho}_{SEZ} = R_2\left(90^\circ - \phi_{gd}\right) R_3\left(\theta_{LST}\right) \vec{\rho}_{IJK}$$

$$\dot{\vec{\rho}}_{SEZ} = R_2\left(90^\circ - \phi_{gd}\right) R_3\left(\theta_{LST}\right) \dot{\vec{\rho}}_{IJK}$$

$$R_{SEZ/IJK} = \begin{bmatrix} \sin\phi_{gd}\cos\theta_{LST} & \sin\phi_{gd}\sin\theta_{LST} & -\cos\phi_{gd} \\ -\sin\theta_{LST} & \cos\theta_{LST} & 0 \\ \cos\phi_{gd}\cos\theta_{LST} & \cos\phi_{gd}\sin\theta_{LST} & \sin\phi_{gd} \end{bmatrix}$$

- $\rho = \left|\vec{\rho}_{SEZ}\right|$
- $\sin\left(el\right) = \dfrac{\rho_Z}{\rho}$
- If $el \neq 90^\circ \iff \rho_s^2 + \rho_E^2 \neq 0$

$$\sin\beta = \frac{\rho_E}{\sqrt{\rho_s^2 + \rho_E^2}} \quad , \quad \cos\beta = \frac{-\rho_S}{\sqrt{\rho_s^2 + \rho_E^2}}$$

If $el = 90^\circ \iff \rho_s^2 + \rho_E^2 = 0$

$$\sin\beta = \frac{\dot{\rho}_E}{\sqrt{\dot{\rho}_s^2 + \dot{\rho}_E^2}} \quad , \quad \cos\beta = \frac{-\dot{\rho}_S}{\sqrt{\dot{\rho}_s^2 + \dot{\rho}_E^2}}$$

- $\dot{\rho} = \dfrac{\vec{\rho}_{SEZ} \cdot \dot{\vec{\rho}}_{SEZ}}{\rho}$
- $$\begin{cases} \vec{\rho}_{SEZ} = \begin{bmatrix} -\rho\cos(el)\cos\beta \\ \rho\cos(el)\sin\beta \\ \rho\sin(el) \end{bmatrix} \\ \dot{\vec{\rho}}_{SEZ} = \begin{bmatrix} -\dot{\rho}\cos(el)\cos\beta + \rho\sin(el)\cos\beta\ \dot{el} + \rho\cos(el)\sin\beta\ \dot{\beta} \\ \dot{\rho}\cos(el)\sin\beta - \rho\sin(el)\sin\beta\ \dot{el} + \rho\cos(el)\cos\beta\ \dot{\beta} \\ \dot{\rho}\sin(el) + \rho\cos(el)\ \dot{el} \end{bmatrix} \end{cases}$$

From the Z-component of $\dot{\vec{\rho}}_{SEZ}$,

$$\dot{el} = \frac{\dot{\rho}_Z - \dot{\rho}\dfrac{\rho_Z}{\rho}}{\sqrt{\rho_S^2 + \rho_E^2}}$$

- $\rho_E\dot{\rho}_S - \rho_S\dot{\rho}_E = \cdots = \dot{\beta}\left(\rho_S^2 + \rho_E^2\right)$

$$\Rightarrow \dot{\beta} = \frac{\rho_E\dot{\rho}_S - \rho_S\dot{\rho}_E}{\rho_S^2 + \rho_E^2}$$

4.4.4. Practical Azimuth-Elevation Conversions

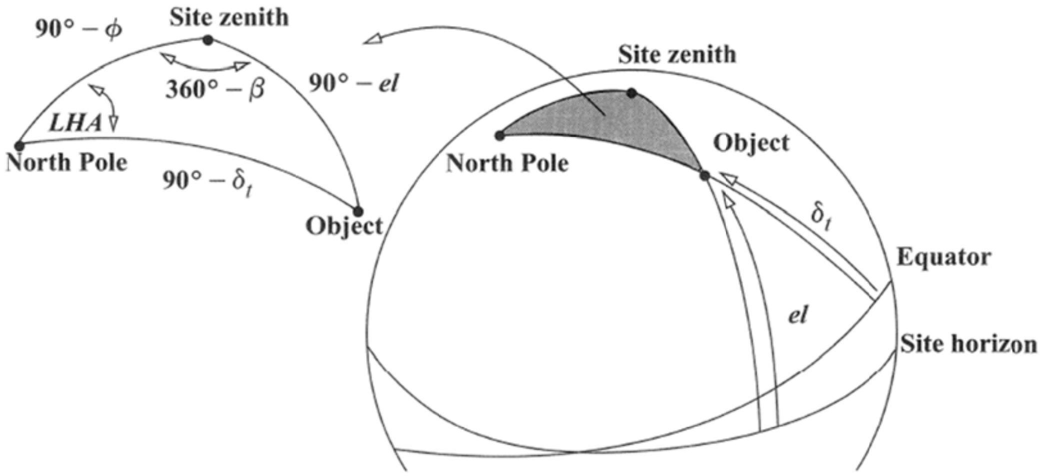


Figure 4-7. **Az-el and RA-dec Geometry.** The spherical triangles shown are necessary to develop relations between the two sets of angles. Don't confuse the site *horizon* with the equator shown in many diagrams.

- Observation Values $\Rightarrow$ State Vectors (in Inertial Frame)

$$\begin{cases} (\rho,\beta,el)_{SEZ} \\ (\rho,\alpha_t,\delta_t)_{I_tJ_tK_t\,(\mathit{Topocentric}\text{ Equatorial Coordination System})} \end{cases} \Rightarrow \vec{r}_{IJK}, \vec{v}_{IJK}$$

  - Form the vectors from the observations

$$\begin{cases} \vec{\rho}_{SEZ} = \begin{bmatrix} -\rho\cos(el)\cos\beta \\ \rho\cos(el)\sin\beta \\ \rho\sin(el) \end{bmatrix} \\ \dot{\vec{\rho}}_{SEZ} = \begin{bmatrix} -\dot{\rho}\cos(el)\cos\beta + \rho\sin(el)\cos\beta\ \dot{el} + \rho\cos(el)\sin\beta\ \dot{\beta} \\ \dot{\rho}\cos(el)\sin\beta - \rho\sin(el)\sin\beta\ \dot{el} + \rho\cos(el)\cos\beta\ \dot{\beta} \\ \dot{\rho}\sin(el) + \rho\cos(el)\ \dot{el} \end{bmatrix} \end{cases}$$

$$\begin{cases} \vec{\rho}_{I_tJ_tK_t} = \begin{bmatrix} \rho\cos\delta\cos\alpha \\ \rho\cos\delta\sin\alpha \\ \rho\sin\delta \end{bmatrix} \\ \dot{\vec{\rho}}_{I_tJ_tK_t} = \begin{bmatrix} \dot{\rho}\cos\delta\cos\alpha - \rho\sin\delta\cos\alpha\ \dot{\delta} - \rho\cos\delta\sin\alpha\ \dot{\alpha} \\ \dot{\rho}\cos\delta\sin\alpha - \rho\sin\delta\sin\alpha\ \dot{\delta} + \rho\cos\delta\cos\alpha\ \dot{\alpha} \\ \dot{\rho}\sin\delta + \rho\cos\delta\ \dot{\delta} \end{bmatrix} \end{cases}$$

- Rotate to Geocentric ECEF frame

$$\begin{cases} \vec{\rho}_{ECEF} = R_{ECEF/SEZ}\vec{\rho}_{SEZ} \\ \vec{\rho}_{ECEF} = R_{ECEF/I_tJ_tK_t}\vec{\rho}_{I_tJ_tK_t} \end{cases}$$

- Translate the origin if necessary

$$\begin{cases} \vec{r}_{ECEF} = \vec{r}_{siteECEF} + \vec{\rho}_{ECEF} \\ \vec{v}_{ECEF} = \dot{\vec{\rho}}_{ECEF} \end{cases}$$

- Perform rotations to Celestial (=Inertial) Frame

$$\begin{cases} \vec{r}_{FK5} = R_{FK5/ECEF}\vec{r}_{ECEF} \\ \vec{v}_{FK5} = R_{FK5/ECEF}\vec{v}_{ECEF} + \dot{R}_{FK5/ECEF}\vec{r}_{ECEF} \end{cases}$$

cf) Refer to sec.3.7 for details

- (Translate the origin to the barycenter if required)
- (Apply corrections as needed)
  - aberration of light
  - relativistic deflection of time
  - light time
  - proper motion
  - parallex
  - radial velocity
- Develop (r,α,δ) from Algorithm 25 if desired

- Conversion between two types of observation comes from spherical geometry

$$(\rho,\beta,el) \;\Rightarrow\; (\rho,\alpha_t,\delta_t)$$

- Algorithm 28 : $\left(\beta, el, \phi_{gc}, \theta_{LST}\right) \Rightarrow \left(\alpha_t, \delta_t\right)$
  - $\sin\delta_t = \sin(el)\sin\phi_{gc} + \cos(el)\cos\phi_{gc}\cos\beta$
  - $\sin LHA = -\dfrac{\sin\beta\cos(el)\cancel{\cos\phi_{gc}}}{\cos\delta_t\cancel{\cos\phi_{gc}}}$

    $\cos LHA = \dfrac{\sin(el) - \sin\phi_{gc}\sin\delta_t}{\cos\delta_t\cos\phi_{gc}}$
  - $\alpha_t = \theta_{LST} - LHA$

4.4.4.1. Transformation for Ecliptic Latitude & Longitude

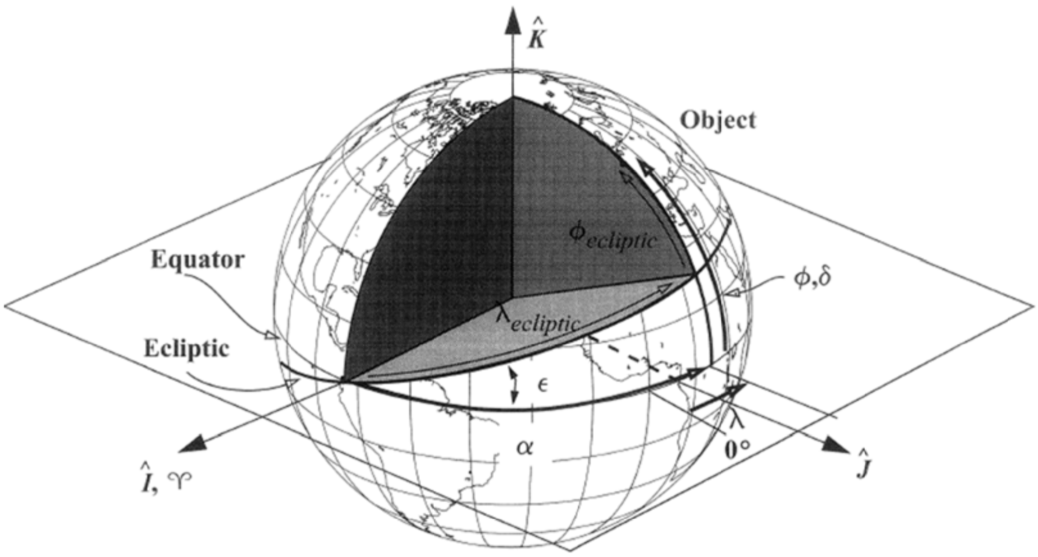


Figure 4-8. Transforming the Ecliptic and Equatorial Systems. Ecliptic coordinates and geocentric coordinates differ by the obliquity of the ecliptic, ϵ and by their origin (heliocentric vs. geocentric, respectively). This permits a simple rotation between systems about the vernal equinox direction.

- Relation between (α, δ) and $(\phi_{ecliptic}, \lambda_{ecliptic})$ is for determining the position of the Sun & Moon, which are common inputs for perturbation analyses
  - Vernal Equinox is common to point to I-axis
  - Geocentric & Ecliptic system differ only by the obliquity of the ecliptic (ε)

- Conversion between (α, δ) and $(\phi_{ec}, \lambda_{ec})$
  - $\vec{r}_{XYZ} = \begin{bmatrix} r\cos\phi_{ec}\cos\lambda_{ec} \\ r\cos\phi_{ec}\sin\lambda_{ec} \\ r\sin\phi_{ec} \end{bmatrix}$
  - $\vec{r}_{IJK} = \mathrm{R}_1(-\varepsilon)\vec{r}_{XYZ}$

$$\Rightarrow \vec{r}_{IJK} = r\begin{bmatrix} \cos\phi_{ec}\cos\lambda_{ec} \\ \cos\varepsilon\cos\phi_{ec}\sin\lambda_{ec} - \sin\varepsilon\sin\phi_{ec} \\ \sin\varepsilon\cos\phi_{ec}\sin\lambda_{ec} + \cos\varepsilon\sin\phi_{ec} \end{bmatrix}$$

  - Unit vectors in both system

$$\hat{L}_{XYZ} = \begin{bmatrix} \cos\phi_{ec}\cos\lambda_{ec} \\ \cos\phi_{ec}\sin\lambda_{ec} \\ \sin\phi_{ec} \end{bmatrix}$$

$$\hat{L}_{IJK} = \begin{bmatrix} \cos\delta\cos\alpha \\ \cos\delta\sin\alpha \\ \sin\delta \end{bmatrix}$$

  - $\hat{L}_{XYZ} = R_1(\varepsilon)\hat{L}_{IJK}$

$$\vdots$$

$$\begin{cases} \sin\phi_{ec} = -\cos\delta\sin\alpha\sin\varepsilon + \sin\delta\cos\varepsilon \\ \sin\lambda_{ec} = \dfrac{\cos\delta\sin\alpha\cos\varepsilon + \sin\delta\sin\varepsilon}{\cos\phi_{ec}} \\ \cos\lambda_{ec} = \dfrac{\cos\delta\cos\alpha}{\cos\phi_{ec}} \end{cases}$$

- $\hat{L}_{IJK} = R_1(-\varepsilon)\hat{L}_{XYZ}$

$$\begin{cases} \sin\delta = \sin\phi_{ec}\cos\varepsilon + \cos\phi_{ec}\sin\varepsilon\sin\lambda_{ec} \\ \sin\alpha = \dfrac{-\sin\phi_{ec}\sin\varepsilon + \cos\phi_{ec}\cos\varepsilon\sin\lambda_{ec}}{\cos\delta} \\ \cos\alpha = \dfrac{\cos\phi_{ec}\cos\lambda_{ec}}{\cos\delta} \end{cases}$$